

Notes: Fang and Peress (JF, 2009)

Media Coverage and the Cross-section of Stock Returns

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1 Big picture

- **Question.** Does mass media coverage affect expected stock returns by broadening the dissemination of information, even when the media is not breaking genuinely new information?
- **Setting.** The paper studies U.S. listed stocks from 1993 to 2002, using all NYSE firms plus 500 randomly selected NASDAQ firms and measuring coverage in four major national newspapers: *Wall Street Journal*, *New York Times*, *Washington Post*, and *USA Today*.
- **Why this question matters.** The media reaches a much broader audience than analyst reports or specialized terminals, especially individual investors. If media coverage increases investor awareness, it can reduce firms' cost of capital even without adding new fundamental information.
- **Core empirical challenge.** Media coverage is strongly correlated with firm size, liquidity, analyst following, and recent public information. A simple return spread between high- and low-coverage firms could just be a disguised size effect, a microstructure artifact, an industry effect, or another known return anomaly.
- **Main conceptual contribution.** The paper interprets the “no-media premium” through two rational-agent channels: *investor recognition* in the spirit of Merton (1987), and *impediments to trade* that keep the premium from being arbitrated away.
- **Main empirical strategy.** Each month, sort stocks into *no coverage*, *low coverage*, and *high coverage* based on the previous month's newspaper article counts. Then test whether a strategy that goes long no-coverage stocks and short high-coverage stocks earns abnormal returns in the next month after controlling for market, size, value, momentum, and liquidity factors.
- **Main baseline result.** Unconditionally, no-coverage stocks outperform high-coverage stocks by about 0.39% per month. After risk adjustment, the long-short strategy still earns about 0.23% per month in the five-factor specification that includes the Pastor–Stambaugh liquidity factor.
- **Key asymmetry.** The alpha comes primarily from the *long leg*: neglected stocks earn positive abnormal returns, while high-coverage stocks do not exhibit statistically significant negative alphas.
- **Key heterogeneity result.** The effect is strongest among small stocks, stocks with low analyst coverage, high individual ownership, high idiosyncratic volatility, and high idiosyncratic volatility per investor. In these groups, the premium is often between roughly 0.65% and 1.1% per month.
- **Main credibility result.** The no-media premium survives controls for standard factors, sorts on size/book-to-market/momentum, returns based on bid-ask midpoints, exclusions of earnings-announcement months and IPOs, corrected delisting returns, exclusion of technology firms, and longer holding horizons up to 12 months.
- **Main interpretation.** The paper does *not* have a sharp causal design. Its strongest conclusion is that the return difference is consistent with mass media affecting investor recognition and information dissemination, while illiquidity helps explain persistence rather than the original source of the premium.

- **Economic takeaway.** Media attention is not just a publicity variable. It can matter for expected returns and therefore for firms' cost of capital because broad dissemination changes who knows about the stock and how well investors can diversify.

2 Data and measurement (key objects)

2.1 Why this setting is useful

- National newspapers are a natural place to test the dissemination channel because they reach many more investors than sell-side analysts or private information networks.
- Newspaper coverage is broad enough to capture investor awareness, but publication delays make it unlikely that newspaper articles are simply delivering private information instantly to the market.
- The sample period, 1993–2002, is early enough that print media still plausibly matters, but late enough to have rich CRSP, Compustat, I/B/E/S, and 13f data for controls and heterogeneity analysis.
- Because the sample contains mostly NYSE stocks and excludes penny stocks, any positive result is conservative: the sample is already tilted toward firms with relatively good information environments.

Interpretation. The paper is not asking whether sensational headlines move prices within minutes. It is asking whether systematic *breadth of coverage* across stocks helps explain the cross-section of expected returns.

2.2 Sample construction and data sources

- Equity sample: all NYSE firms plus 500 randomly selected NASDAQ firms.
- Sample period: January 1993 to December 2002.
- Keep only stocks with prices above \$5, following standard practice to reduce microstructure distortions and very illiquid tails.
- Media data: LexisNexis search results for four national dailies — *WSJ*, *NYT*, *WP*, and *USA Today*.
- For each company, the authors use LexisNexis indexing keywords and retain only articles with a relevance score of at least 90% (“Major References”).
- Return, price, market cap, and volume data come from CRSP.
- Accounting data come from Compustat.
- Analyst coverage comes from I/B/E/S summary files.
- Institutional ownership comes from aggregated 13f filings; the fraction of individual ownership is calculated as one minus institutional ownership.

Interpretation. The data architecture is built to connect three objects: media exposure, standard return predictors, and proxies for both liquidity and information incompleteness.

2.3 How media coverage is measured

For firm i in month t , the paper’s main media variable is the circulation-weighted count of newspaper articles:

$$\text{Media}_{i,t} = \sum_{n=1}^4 w_n \cdot \text{Articles}_{i,t}^{(n)},$$

where n indexes the four newspapers and w_n is the newspaper’s circulation weight.

Each month, firms are sorted into three groups:

- **No coverage:** zero articles in the previous month.
- **Low coverage:** positive coverage, but article count at or below the median among covered firms.
- **High coverage:** positive coverage above the median among covered firms.

Interpretation. This is a measure of *visibility* or *recognition*, not of tone or sentiment. The paper is about whether being in or out of the broad public media matters, not whether the media is positive or negative.

2.4 Dependent variables and portfolio objects

The main outcome is next month’s stock return, organized through portfolio sorts.

- The basic raw-return object is the equal-weighted average return in month $t+1$ for each coverage portfolio formed using month- t coverage.
- The main abnormal-return object is the return on a zero-investment strategy that goes long no-coverage stocks and short high-coverage stocks.
- The paper then estimates alphas from CAPM, Fama–French three-factor, Carhart four-factor, and Pastor–Stambaugh five-factor models.
- In additional exercises, the paper computes DGTW characteristic-adjusted excess returns and horizon returns using overlapping portfolios.

Interpretation. The return question is not “do news stories move prices today?” It is “do low-visibility stocks earn higher returns next month and beyond?”

2.5 Controls, liquidity proxies, and information proxies

The paper uses several firm characteristics to describe where the media effect is strongest.

Standard return predictors.

- **Size:** log of average market capitalization over the previous calendar year.

- **Book-to-market:** log of book equity divided by market equity at the previous year-end.
- **Past return / momentum:** signed or absolute past month or past year return.

Liquidity / impediments-to-trade proxies.

- Amihud (2002) illiquidity ratio.
- Bid-ask spread.
- Dollar trading volume.
- Stock price.

Investor-recognition / information-incompleteness proxies.

- Analyst coverage.
- Fraction of individual ownership.
- Analyst forecast dispersion.
- Idiosyncratic volatility from daily residuals relative to a Fama–French three-factor model.
- Idiosyncratic volatility per investor.

Interpretation. The paper’s basic logic is: if media matters through awareness, the media premium should be stronger exactly where other information channels are weak or where undiversified risk is more costly.

2.6 Key descriptive facts

- Overall newspaper coverage is much lower than one might guess. Across all years, 70% of sample firms receive coverage in a typical year.
- Coverage is higher for NYSE firms (73%) than for NASDAQ firms (42%).
- Conditional on being covered, the average stock gets about 12 articles per year, while the median gets only 5.
- Coverage is therefore highly skewed: a small set of firms receives a lot of attention, while a sizable group is effectively ignored.
- Coverage is persistent: 83% of no-coverage stocks in one month remain uncovered in the next month, and 49% of high-coverage stocks remain high-coverage in the next month.
- Figure 1 shows that no-coverage and covered stocks have almost identical one-digit SIC distributions, which helps rule out a simple industry-composition story.

The determinants regression further shows that:

- size is the dominant positive predictor of coverage;
- book-to-market is positive conditional on size;
- analyst coverage enters negatively, suggesting media and analyst coverage may be substitutes at the margin;

- individual ownership, analyst dispersion, and idiosyncratic volatility are all positively related to coverage;
- past signed and absolute returns do not materially predict coverage.

Interpretation. Media coverage is a persistent firm characteristic, strongly related to size and information environment, but not obviously driven by a particular industry or by simple return-chasing.

3 Models and empirical specifications (all equations + interpretation)

3.1 Economic mechanism and hypotheses

The paper develops two rational-agent interpretations of a return premium for neglected stocks.

1. Investor recognition hypothesis.

- Following Merton (1987), if investors do not know all securities, then poorly recognized stocks are held by fewer investors.
- Those investors are less diversified and therefore require higher expected returns.
- Media coverage broadens awareness and increases investor recognition, so better-covered stocks should earn lower expected returns.

2. Impediments-to-trade hypothesis.

- If the no-media premium is a mispricing, it can survive only if frictions such as illiquidity or trading costs prevent arbitrage.
- This implies the media premium should be stronger where trading is more difficult.

These hypotheses generate a common headline prediction — neglected stocks earn higher expected returns — but different cross-sectional implications about where the effect should be strongest.

Interpretation. Investor recognition explains *why* the premium may arise; impediments to trade explain *why it may persist*. The paper ultimately leans toward this two-step view.

3.2 Univariate portfolio sorts

The basic raw-return test is:

$$R_{t+1}^{LS} = R_{t+1}^{\text{No}} - R_{t+1}^{\text{High}},$$

where R_{t+1}^{No} is the equal-weighted next-month return on stocks with zero coverage in month t , and R_{t+1}^{High} is the equal-weighted next-month return on stocks with above-median coverage in month t .

The paper also performs *double sorts*. First sort stocks into terciles on a characteristic such as size, price, or momentum; then within each tercile, split into no/low/high media groups and compare next-month returns.

Interpretation. This design shows whether the no-media premium survives controlling for one characteristic at a time in a transparent way.

3.3 Factor-model time-series regressions

The main abnormal-return specification is a time-series regression of the long-short portfolio return on standard factors:

$$R_t^{LS} = \alpha + \beta_M(Mkt - rf)_t + \beta_S SMB_t + \beta_H HML_t + \beta_U UMD_t + \beta_L LIQ_t + \varepsilon_t,$$

with subsets of the factors used in four nested models:

- **Model 1:** CAPM.
- **Model 2:** Fama–French three-factor model.
- **Model 3:** Carhart four-factor model.
- **Model 4:** Pastor–Stambaugh liquidity-augmented specification.

The parameter of interest is α .

Interpretation. If $\alpha > 0$, then neglected stocks outperform well-covered stocks by more than can be explained by standard factor exposures.

3.4 Coverage-determinants regressions

To understand what kinds of firms receive attention, the paper estimates annual Fama–MacBeth regressions of the form

$$\text{Media}_{i,t} = a_t + b_{1,t} \text{Size}_{i,t} + b_{2,t} \text{B/M}_{i,t} + b_{3,t} \text{AnalystCov}_{i,t} + b_{4,t} \text{IndivOwn}_{i,t} + b_{5,t} \text{Dispersion}_{i,t} + b_{6,t} \text{IdioVol}_{i,t} + u_{i,t},$$

with variants that also include past returns.

The reported coefficients are time-series averages of the cross-sectional slopes, and the paper uses Newey–West corrections with one lag because media coverage is persistent.

Interpretation. These regressions do not identify a causal effect of coverage. They simply map media attention to observable firm characteristics and help clarify what must be controlled for in return tests.

3.5 Horizon analysis

To distinguish a stable cross-sectional premium from a short-lived reversal, the paper forms portfolios using coverage over the past 1, 3, or 6 months and then studies post-formation returns for holding periods from 1 to 12 months using calendar-time overlapping portfolios.

Interpretation. If the no-media premium were just a one-month reversal, it should quickly disappear. Persistence over many months is more consistent with a stable difference in expected returns.

3.6 Liquidity and investor-recognition heterogeneity

The long–short alpha is recomputed within terciles sorted by:

- liquidity proxies: Amihud illiquidity, bid-ask spread, dollar volume, price;
- investor-recognition proxies: analyst coverage, fraction of individual ownership, idiosyncratic volatility, idiosyncratic volatility per investor.

The paper also uses DGTW characteristic-adjusted returns in double-sorts with idiosyncratic volatility and analyst dispersion.

Interpretation. These tests ask two sharper questions: does the media premium live where arbitrage is hardest, and does it live where awareness is poorest?

4 Identification and instruments (what makes the estimates credible)

4.1 What identifies the baseline effect

The paper identifies the baseline relation from cross-sectional variation in lagged media coverage across publicly listed stocks and asks whether that variation predicts next-month returns.

The key features are:

- coverage is measured *before* the return is realized;
- known factor exposures are partialled out with standard asset-pricing models;
- characteristic sorts make sure the premium is not simply one-for-one a size, value, momentum, or price effect;
- the long–short strategy is rebalanced monthly, so the estimate uses repeated cross-sectional comparisons rather than a single episode.

Interpretation. The design identifies a strong *predictive association* between coverage and future returns. It is not a quasi-experiment.

4.2 No formal instrument and no causal shock

This paper does **not** use an instrumental variable, a natural experiment, or an exogenous media shock. There is no attempt to claim that media coverage is randomly assigned.

Instead, credibility comes from pattern-based evidence:

- the alpha survives many standard risk controls;
- it survives several well-known anomaly-based alternative explanations;
- it is strongest in subsamples where theory says media should matter more;
- it is stable over longer horizons, which is hard to reconcile with pure short-term reversal.

Interpretation. The section title in the template says “Identification and instruments,” but for this paper the honest statement is: there is no instrument. The evidence is suggestive and carefully stress-tested rather than cleanly causal.

4.3 Why the effect is not obviously a disguised industry or technology result

- Figure 1 shows that covered and uncovered stocks have nearly identical one-digit SIC distributions.
- Excluding technology-sector stocks leaves the premium intact.

Interpretation. The no-media premium is not just “buy non-tech, short tech” in disguise.

4.4 Why the effect is not obviously a microstructure artifact

- The sample already excludes stocks priced below \$5.
- The sample is mostly NYSE stocks, which are much more liquid than the full CRSP universe.
- Recomputing returns using bid-ask midpoints rather than transaction prices leaves the alpha very similar and slightly stronger.

Interpretation. Bid-ask bounce and thin trading are unlikely to be the main explanation.

4.5 Why the effect is distinct from return drift and reversal stories

A central concern is Chan (2003): perhaps the results merely reflect drift for “news losers” or reversal for “no-news losers.” The paper addresses this in three ways.

- **Coverage is not the same as news.** The authors report that 92% of high-coverage stocks have contemporaneous headline news, but so do 76% of no-coverage stocks. Many firms with genuine news still remain neglected by the mass media.
- **The alpha comes from the long leg.** If the result came from negative drift in high-coverage losers, the short leg should drive the alpha. Instead, abnormal returns come mainly from the no-media leg.
- **The effect is long-lived.** Table VII and Figure 2 show that the premium remains visible for holding periods up to 12 months, especially with longer formation windows, unlike a short-lived reversal.

Interpretation. The paper does a good job showing that “low media coverage” is not just a relabeling of Chan’s “no-news” category.

4.6 What the paper can and cannot distinguish

What it can establish fairly well.

- A robust negative relation exists between media coverage and future returns.
- The relation is concentrated where investor recognition is likely to be low.
- Some liquidity-based evidence is consistent with arbitrage frictions helping the premium persist.

What it cannot pin down cleanly.

- It cannot cleanly separate risk compensation from mispricing.
- It cannot prove that media coverage itself, rather than some omitted but correlated awareness variable, is the causal object.
- It does not decisively arbitrate between investor recognition and impediments-to-trade, though it argues the former is the deeper explanation.
- It does not provide a structural model that maps media exposure into a quantitative change in cost of capital.

4.7 Relation to the literature

The paper sits at the intersection of four literatures:

- media and asset prices;
- investor recognition and incomplete-information asset pricing;
- return anomalies linked to news, analyst dispersion, and idiosyncratic volatility;
- liquidity and limits-to-arbitrage.

Its main contribution is to turn “media coverage” into a cross-sectional firm characteristic with pricing content, not just a short-run event variable.

5 Tables and figures

5.1 Figure 1: industry composition does not drive the result

Read:

- The industry histograms for covered and uncovered stocks are almost identical.
- Both groups are spread across the same broad one-digit SIC sectors.

Economic message. The no-media premium is not just an industry-composition artifact. That matters because otherwise the return spread could simply reflect omitted industry risk or the tech boom.

5.2 Table I: media coverage is low, skewed, and persistent

Read:

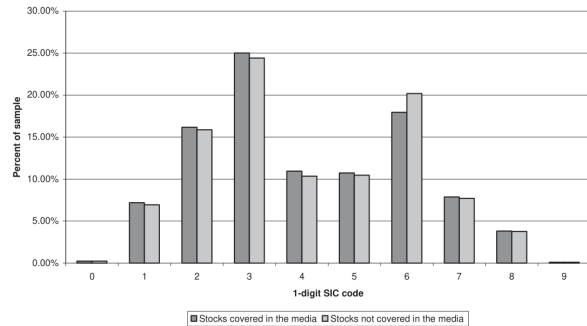


Figure 1. Industry distribution of media coverage. The histogram shows the industry distribution of stocks covered by the media and of stocks not covered by the media. The one-digit SIC classification is as follows—0: agriculture, forestry, and fishing; 1: mining and construction; 2: manufacturing (consumer goods); 3: manufacturing (machinery and equipment); 4: transportation and communications; 5: wholesale and retail; 6: finance; 7: business services; 8: health and education services; 9: public administration.

Table I
Summary Statistics of Newspaper Coverage

This table presents summary statistics for the newspaper coverage of our sample firms. Both unconditional statistics (percentage of firms receiving coverage) and conditional statistics (number of articles written on the firm conditioned on coverage) are presented. The column “All papers” refers to all four national newspapers in our sample: *Wall Street Journal* (WSJ), *New York Times* (NYT), *Washington Post* (WP), and *USA Today* (USAT). The column “Excl. WSJ” represents the three nonfinancial papers: NYT, WP, and USAT.

Year	Unconditional Coverage Statistics Fraction of Stocks Covered by						Conditional Statistics No. of Articles	
	All Papers	WSJ	Excl. WSJ	NYT	WP	USAT	Mean	Median
Panel A: All Stocks								
1993	0.77	0.61	0.60	0.58	0.12	0.07	11	4
1994	0.75	0.60	0.59	0.58	0.12	0.07	12	5
1995	0.75	0.61	0.60	0.59	0.11	0.07	11	5
1996	0.72	0.56	0.60	0.59	0.12	0.06	11	5
1997	0.73	0.56	0.61	0.59	0.13	0.06	11	5
1998	0.75	0.59	0.64	0.62	0.15	0.06	11	5
1999	0.68	0.58	0.50	0.47	0.15	0.04	12	4
2000	0.63	0.52	0.49	0.46	0.14	0.05	12	4
2001	0.62	0.52	0.44	0.42	0.14	0.05	12	4
2002	0.57	0.46	0.44	0.41	0.16	0.04	12	4
All years	0.70	0.57	0.56	0.54	0.13	0.06	12	5
Panel B: NYSE Stocks								
All years	0.73	0.59	0.59	0.57	0.14	0.06	12	5
Panel C: NASDAQ Stocks								
All Years	0.42	0.31	0.27	0.24	0.05	0.02	4.2	2

- Only 70% of sample stocks are covered in a typical year; among NASDAQ stocks the fraction is only 42%.
- Conditional on coverage, the mean number of articles is about 12 but the median is only 5.
- The descriptive discussion after the table shows strong persistence in coverage status from month to month.

Economic message. “No coverage” is not a rare tail event. It is a meaningful and persistent state for many listed firms, especially smaller ones.

5.3 Table II: what kinds of firms get covered?

Read:

- Size has by far the largest positive relation with coverage.
- Conditional on size, book-to-market is positive.
- Analyst coverage is negative, suggesting media coverage and analyst coverage are not simple complements.
- Individual ownership, analyst dispersion, and idiosyncratic volatility all enter positively.
- Past returns are not important predictors of coverage.

Economic message. The paper has to control hard for firm characteristics because the media does not cover stocks randomly. Table II tells us exactly which confounds are most important.

5.4 Table III: the raw no-media premium is economically large

Read:

Table II
Determinants of Media Coverage

This table reports Fama-Macbeth (1973) regression results on the determinants of media coverage. The dependent variable is the number of articles published about a stock in a given year. Independent variables are defined in Table AI. *t*-statistics are based on standard errors adjusted for autocorrelation using the Newey-West (1987) procedure with one lag. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% level, respectively.

Dependent Variable: Number of Articles		
Size	1.084 (20.17)**	1.081 (19.62)**
Book-to-market	0.233 (12.06)**	0.228 (11.98)**
Analyst coverage	-0.518 (6.93)**	-0.51 (6.86)**
Fraction of individual ownership	0.183 (3.79)**	0.177 (3.77)**
Analyst dispersion	0.223 (3.09)*	0.212 (2.50)*
Idiosyncratic volatility	42.364 (6.35)**	42.668 (7.03)**
Past year absolute return	0.009 -0.08	
Past year return		-0.081 -1.04
Constant	-14.167 (23.91)**	-14.117 (23.54)**
Observations	13849.00	13849.00
<i>R</i> ²	0.24	0.24

Table III
Media Coverage and Stock Returns: Univariate Comparisons

This table presents average monthly returns for stocks with no, low, and high newspaper coverage. Average return numbers are in percentages. Each month, we divide our sample of firms into three media-coverage portfolios: no coverage, low coverage, and high coverage. Media coverage is measured by the number of newspaper articles written about the company, and the median is used to divide the covered stocks into low and high groups. We then compute the equal-weighted average return of the three media coverage portfolios using individual stock returns in the next month. We also compute the return difference for subsamples of firms sorted on size, book-to-market ratio, current and past month returns, price, individual ownership, analyst coverage, illiquidity, and turnover. These variables are defined in Table AI.

	Average Monthly Return					Average No. of Stocks		
	Media Coverage			<i>t</i> -Statistics for		Media Coverage		
	No	Low	High	No – High	No – High	No	Low	High
All stocks	1.35	1.11	0.96	0.39	2.13	1,430.08	284.82	245.40
Panel A: By Size								
1	1.41	1.02	0.53	0.88	1.74	578.44	56.36	17.98
2	1.34	1.12	0.69	0.65	2.68	514.23	92.85	46.71
3	1.27	1.16	1.10	0.17	1.03	337.42	167.12	149.19
Panel B: By Book-to-Market								
1	1.19	0.95	0.87	0.32	1.25	441.79	93.98	81.50
2	1.23	1.17	0.54	0.70	3.13	450.03	92.90	74.64
3	1.42	1.10	1.19	0.22	0.85	460.78	86.20	70.57
Panel C: By Past Month Return								
1	1.43	0.98	0.85	0.58	2.29	474.54	93.31	76.78
2	1.25	0.89	0.91	0.34	1.73	461.16	97.56	82.13
3	1.09	1.07	0.64	0.44	2.19	467.33	93.25	78.77
Panel D: By Current Month Return								
1	1.96	1.49	0.86	1.10	4.24	479.50	94.11	75.77
2	1.29	1.08	1.01	0.28	1.36	475.29	94.21	79.54
3	0.88	0.75	1.08	-0.20	-0.76	467.93	100.31	84.40
Panel E: By Price								
1	1.01	0.53	-0.11	1.11	3.14	545.18	69.76	35.76
2	1.39	1.12	0.54	0.84	3.77	500.77	94.03	60.14
3	1.77	1.47	1.35	0.42	2.62	384.13	142.30	128.21

- Unconditionally, no-coverage stocks earn 1.35% per month versus 0.96% for high-coverage stocks; the raw spread is 0.39% per month.
- The spread is especially large among small stocks, low-price stocks, and loser stocks based on the current month.
- With only one exception, the no-minus-high difference is positive across the displayed double sorts.

Economic message. Before any factor adjustment, the return spread is already economically meaningful. The rest of the paper is about whether this survives risk adjustment and what it means.

5.5 Table IV: the baseline alpha survives standard risk adjustments

Read:

- The long-short strategy alpha is 0.45% per month in CAPM, 0.35% in Fama-French, 0.24% in Carhart, and 0.23% in the Pastor-Stambaugh liquidity model.
- The strategy loads positively on SMB, HML, and UMD, and negatively on the market factor.
- Panel B shows that no-media stocks themselves earn positive alphas; Panel C shows high-media stocks do not have significant alphas.

Economic message. The no-media premium is not wiped out by standard asset-pricing controls, and it is driven by neglected stocks earning unusually high returns rather than by glamour stocks crashing.

Table IV

Media-Related Trading Profits: Baseline Multivariate Results

This table examines the profitability of a trading strategy that longs stocks with no media coverage and shorts stocks with high media coverage. Each month, stocks are sorted according to the number of newspaper articles published about them. A stock is considered to have no media coverage if no article is published about the stock in a given month. A stock is considered to have high coverage if the number of articles about it exceeds the medium in a given month. Both the long and short positions are equally weighted, and held for 1 month after portfolio formation. Portfolios are rebalanced monthly. The resulting time-series returns on the long-short portfolio are regressed on widely accepted risk factors (defined in Table A1), and the results are reported. p -values are in parentheses. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% level, respectively.

	Model 1: CAPM	Model 2: FF Three-Factor	Model 3: Carhart Four-Factor	Model 4: PS Liquidity
Panel A: Long No-Media Stocks, Short High-Media Stocks				
Mkt- r_f	-0.1434*** (0.0002)	-0.1182*** (0.0004)	-0.0910*** (0.0053)	-0.0918*** (0.0050)
SMB	-	0.3719*** (0.0000)	0.3565*** (0.0000)	0.3602*** (0.0000)
HML	-	0.1580*** (0.0004)	0.1732*** (0.0001)	0.1620*** (0.0003)
UMD	-	-	0.0767*** (0.0006)	0.0939*** (0.0017)
LIQ	-	-	-	-2.5419 (0.3783)
Intercept	0.0045** (0.0110)	0.0035*** (0.0051)	0.0024** (0.0471)	0.0023* (0.0611)
Observations	119	119	119	119
R^2	0.11	0.58	0.62	0.62
Panel B: Alphas for No-Media Coverage Stocks				
Intercept	0.0065*** (0.0072)	0.0024 (0.1020)	0.0042*** (0.0023)	0.0039*** (0.0047)
Panel C: Alphas for High-Media Coverage Stocks				
Intercept	0.002 (0.3263)	-0.0011 (0.4749)	0.0018 (0.1263)	0.0016 (0.1859)

Table VI

Robustness Checks

This table reports returns of a long-short portfolio that goes long on stocks with no media coverage in the previous month and goes short on stocks with high (above median) media coverage in the previous month, after applying various data screens. The long and short legs of the portfolio invest an equal amount in each underlying stock. p -values are in parentheses. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% level, respectively.

	Model 1: CAPM	Model 2: FF Three-Factor	Model 3: Carhart Four-Factor	Model 4: PS Liquidity Factor
Panel A: Returns Based on Bid-Ask Midpoints				
Intercept	0.0051*** (0.0051)	0.0043*** (0.0010)	0.0030** (0.0159)	0.0029** (0.0195)
Panel B: Excluding Earnings Announcements				
Intercept	0.0056*** (0.0017)	0.0049*** (0.0035)	0.0046*** (0.0082)	0.0046*** (0.0080)
Panel C: Excluding IPOs				
Intercept	0.0046*** (0.0086)	0.0033*** (0.0081)	0.0021* (0.0789)	0.002 (0.1036)
Panel D: With Corrected Delisting Returns				
Intercept	0.0045** (0.0109)	0.0035*** (0.0047)	0.0024** (0.0470)	0.0023* (0.0607)
Panel E: Applying All Three Filters (B-D)				
Intercept	0.0053*** (0.0038)	0.0044** (0.0133)	0.0039** (0.0332)	0.0039** (0.0343)
Panel F: Excluding Tech-Sector Stocks				
Intercept	0.0038** (0.0264)	0.0035*** (0.0022)	0.0031*** (0.0086)	0.0031*** (0.0097)

5.6 Table VI: the result survives obvious alternative explanations

Read:

- Using bid-ask midpoint returns leaves the alpha intact.
- Excluding earnings-announcement months actually strengthens the alphas.
- Excluding IPOs and correcting delisting returns leave the results largely unchanged.
- Excluding technology stocks still produces a strong positive alpha.

Economic message. The result is not just post-earnings drift, IPO underperformance, delisting bias, bid-ask bounce, or a pure tech-sector phenomenon.

5.7 Table VII and Figure 2: the effect is stable over long horizons

Read:

- The long-short alpha remains positive for holding periods up to 12 months.
- Longer formation windows of 3 or 6 months make the premium even more stable.
- Figure 2 shows the alpha profile of the long and short legs separately: the long no-media leg drives the effect throughout the horizon.

Economic message. This is hard to reconcile with a one-month reversal or a mechanical no-news bounce. The premium behaves more like a stable expected-return difference.

Table VII
Different Formation Periods and Holding Periods

This table reports mean returns for the long–short portfolio that goes long on stocks with no media coverage over the past N months (the “formation period”) and short on stocks with high (above-median) media coverage over the past N months ($N = 1, 3, 6$). Average monthly alphas for various holding horizons between 1 month and 12 months are reported (the “holding period”). *, **, and *** indicate statistical significance at the 10%, 5%, and 1% level, respectively.

Holding Period	Time-Series Mean	CAPM Alpha	FF Three-Factor Alpha	Carhart Four-Factor Alpha	PS Five-Factor Alpha
Panel A: Formation Period = 1 Month					
1 month	0.0039**	0.0045**	0.0035***	0.0024**	0.0023*
3 months	0.0033*	0.0039**	0.0028**	0.0016	0.0015
6 months	0.003	0.0036**	0.0026**	0.0013	0.0012
9 months	0.0026	0.0033*	0.0024**	0.0012	0.0011
12 months	0.0027	0.0034*	0.0025**	0.0013	0.0011
Panel B: Formation Period = 3 Months					
1 month	0.0036**	0.0042**	0.0033***	0.0022**	0.0020*
3 months	0.0032*	0.0038**	0.0030***	0.0017	0.0015
6 months	0.0029	0.0035**	0.0028**	0.0016	0.0015
9 months	0.0028	0.0035**	0.0029***	0.0017*	0.0015
12 months	0.0028	0.0034*	0.0030***	0.0017*	0.0016
Panel C: Formation Period = 6 Months					
1 month	0.0032*	0.0038**	0.0032***	0.0022**	0.0021**
3 months	0.0029	0.0034**	0.0030***	0.0021**	0.0020**
6 months	0.0027	0.0033*	0.0030***	0.0021**	0.0020*
9 months	0.0027	0.0033*	0.0032***	0.0022**	0.0021**
12 months	0.0028	0.0035*	0.0033***	0.0022**	0.0021*

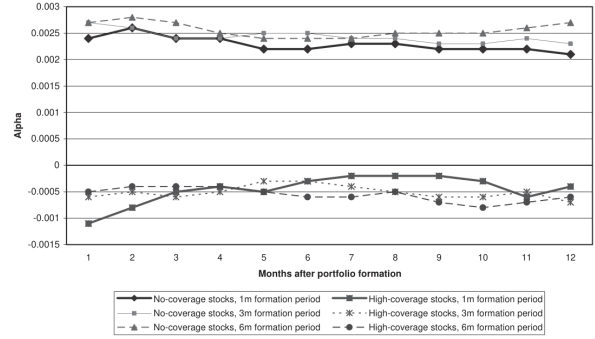


Figure 2. Horizon analysis of the media effect. Fama–French three-factor adjusted alphas for no- and high-coverage stocks are displayed for various formation and holding periods. Stocks are assigned to portfolios based on their coverage in the media over the past 1, 3, or 6 months. The portfolio returns are plotted for holding horizons ranging from 1 month to 12 months.

Table VIII
Illiquidity and the Media Effect

This table examines the profitability of a media-based trading strategy in subsamples of firms sorted by various liquidity measures. Monthly alphas from various factor models of a long–short strategy that goes long no-coverage stocks and short high-coverage stocks in the previous month are reported. Equal weights are used in each leg, and portfolios are rebalanced monthly. The liquidity measures are defined in Table AI. p -values are in parentheses. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% level, respectively.

CAPM	FF Three-Factor	Carhart Four-Factor	PS Liquidity
Panel A: By Amihud's (2002) Illiquidity Ratio			
Low			
0.0025 (0.1162)	0.0016 (0.3202)	−0.0004 (0.7609)	−0.0005 (0.7281)
Medium			
0.0113*** (0.0001)	0.0119*** (0.0000)	0.0097*** (0.0007)	0.0093*** (0.0012)
High			
0.0046 (0.3084)	0.0035 (0.4574)	0.0021 (0.6603)	0.0022 (0.6477)
Panel B: By Bid-Ask Spread			
Low			
0.0001 (0.9433)	−0.001 (0.5113)	−0.0012 (0.4694)	−0.0012 (0.4781)
Medium			
0.0096*** (0.0000)	0.0084*** (0.0000)	0.0074*** (0.0003)	0.0071*** (0.0005)
High			
0.0098*** (0.0010)	0.0096*** (0.0009)	0.0086*** (0.0039)	0.0095*** (0.0010)
Panel C: By Dollar Trading Volume			
Low			
0.0063 (0.1815)	0.0089* (0.0641)	0.0066 (0.1802)	0.0056 (0.2494)
Medium			
0.0090*** (0.0035)	0.0084*** (0.0077)	0.0070** (0.0278)	0.0073** (0.0240)
High			
0.0047** (0.0129)	0.0041** (0.0243)	0.0023 (0.1805)	0.0022 (0.2056)
Panel D: By Price			
Low			
0.0128*** (0.0001)	0.0132*** (0.0001)	0.0099*** (0.0015)	0.0101*** (0.0014)
Medium			
0.0090*** (0.0001)	0.0083*** (0.0002)	0.0054*** (0.0065)	0.0053*** (0.0084)
High			
0.0045*** (0.0049)	0.0035*** (0.0067)	0.0024* (0.0595)	0.0023* (0.0747)

5.8 Table VIII: liquidity helps, but the evidence is mixed

Read:

- The media premium is strongest in high bid-ask spread stocks and in low-price stocks.
- But the relation is not monotonic in every liquidity proxy: for Amihud illiquidity and dollar volume, the effect is strongest in the middle tercile rather than the extreme illiquid tercile.

Economic message. Illiquidity probably helps the premium persist, but the evidence is not clean enough to say that limits to arbitrage are the whole story.

5.9 Table IX: investor recognition is where the paper is strongest

Table IX
Investor Recognition and the Media Effect

This table examines the profitability of a media-based trading strategy in firms sorted by investor recognition. Monthly alphas from various factor models of a long-short strategy that goes long no-coverage stocks and short high-coverage stocks in the previous month are reported. Equal weights are used in each leg, and portfolios are rebalanced monthly. The investor recognition measures are defined in Table A1. *p*-values are in parentheses. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% level, respectively.

CAPM	FF Three-Factor	Carhart Four-Factor	PS Liquidity
Panel A: By Analyst Coverage			
	No		
0.0069 (0.1142)	0.0072 (0.1007)	0.0078* (0.0866)	0.0082* (0.0748)
	Low		
0.0081*** (0.0001)	0.0077*** (0.0000)	0.0070*** (0.0014)	0.0067*** (0.0020)
	High		
0.0015 (0.4048)	-0.0001 (0.9701)	-0.0015 (0.3639)	-0.0018 (0.2736)
Panel B: By the Fraction of Individual Ownership			
	Low		
0.0025 (0.2121)	0.0011 (0.4586)	0.0001 (0.9371)	0 (0.9997)
	Medium		
0.0061*** (0.0078)	0.0042** (0.0200)	0.0024 (0.1629)	0.0021 (0.2383)
	High		
0.0089*** (0.0014)	0.0101*** (0.0001)	0.0094*** (0.0000)	0.0093*** (0.0000)
Panel C: By Idiosyncratic Volatility			
	Low		
0.0000 (0.9984)	-0.0002 (0.8557)	0.0002 (0.8837)	0.0008 (0.5644)
	Medium		
0.0040** (0.0349)	0.0022 (0.1324)	0.0016 (0.2959)	0.0015 (0.3205)
	High		
0.0090*** (0.0018)	0.0090*** (0.0034)	0.0060** (0.0421)	0.0062** (0.0359)
Panel D: By Idiosyncratic Volatility per Investor			
	Low		
0.0018 (0.2447)	-0.0002 (0.8612)	-0.0014 (0.2838)	-0.0018 (0.1774)
	Medium		
0.0056* (0.0654)	0.0060* (0.0538)	0.0024 (0.4172)	0.0025 (0.3913)
	High		
0.0108*** (0.0029)	0.0112*** (0.0029)	0.0087** (0.0199)	0.0077** (0.0362)

Table IX
Investor Recognition and the Media Effect

This table examines the profitability of a media-based trading strategy in firms sorted by investor recognition. Monthly alphas from various factor models of a long-short strategy that goes long no-coverage stocks and short high-coverage stocks in the previous month are reported. Equal weights are used in each leg, and portfolios are rebalanced monthly. The investor recognition measures are defined in Table A1. *p*-values are in parentheses. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% level, respectively.

CAPM	FF Three-Factor	Carhart Four-Factor	PS Liquidity
Panel A: By Analyst Coverage			
	No		
0.0069 (0.1142)	0.0072 (0.1007)	0.0078* (0.0866)	0.0082* (0.0748)
	Low		
0.0081*** (0.0001)	0.0077*** (0.0000)	0.0070*** (0.0014)	0.0067*** (0.0020)
	High		
0.0015 (0.4048)	-0.0001 (0.9701)	-0.0015 (0.3639)	-0.0018 (0.2736)
Panel B: By the Fraction of Individual Ownership			
	Low		
0.0025 (0.2121)	0.0011 (0.4586)	0.0001 (0.9371)	0 (0.9997)
	Medium		
0.0061*** (0.0078)	0.0042** (0.0200)	0.0024 (0.1629)	0.0021 (0.2383)
	High		
0.0089*** (0.0014)	0.0101*** (0.0001)	0.0094*** (0.0000)	0.0093*** (0.0000)
Panel C: By Idiosyncratic Volatility			
	Low		
0.0000 (0.9984)	-0.0002 (0.8557)	0.0002 (0.8837)	0.0008 (0.5644)
	Medium		
0.0040** (0.0349)	0.0022 (0.1324)	0.0016 (0.2959)	0.0015 (0.3205)
	High		
0.0090*** (0.0018)	0.0090*** (0.0034)	0.0060** (0.0421)	0.0062** (0.0359)
Panel D: By Idiosyncratic Volatility per Investor			
	Low		
0.0018 (0.2447)	-0.0002 (0.8612)	-0.0014 (0.2838)	-0.0018 (0.1774)
	Medium		
0.0056* (0.0654)	0.0060* (0.0538)	0.0024 (0.4172)	0.0025 (0.3913)
	High		
0.0108*** (0.0029)	0.0112*** (0.0029)	0.0087** (0.0199)	0.0077** (0.0362)

hypothesis. Panels A and B show that the media effect is stronger among stocks with low analyst coverage and a high percentage of individual ownership. These stocks are poorly covered by conventional information channels, and our results suggest that media coverage plays a large incremental role. Panels C and D show that the magnitude and significance of the media effect monotonically increase with idiosyncratic volatility and idiosyncratic volatility per shareholder, consistent with the predictions of the Merton model. The magnitude of the effect is about 1% per month among stocks with the highest measures (Panels C and D), which is economically large.

In summary, the results in the last two sections provide support for both the illiquidity hypothesis and the investor recognition hypothesis. However, although illiquidity may explain why the media effect persists, it does not explain why it arises in the first place. We conclude that the media effect may stem from media's role in enhancing investor recognition, and that a lack of adequate liquidity helps explain why it is not arbitrated away.

D. Media, Analyst Dispersion, and Idiosyncratic Volatility

An interesting question is whether the media effect is subsumed under recently documented anomalies related to analyst dispersion and idiosyncratic volatility. Diether et al. (2002) (DMS) document that stocks with high analyst forecast dispersion exhibit low future returns. Building on Miller (1977), they argue that forecast dispersion proxies for heterogeneity in investors' opinions and that under short-sales constraints, prices reflect the most optimistic views. Ang et al. (2006) document that stocks with high idiosyncratic volatility earn low future returns. This is at odds with the notion that investors should be rewarded for bearing risk that they cannot diversify away (e.g., Merton (1987)). Furthermore, this effect cannot be explained by stock returns to systematic

Read:

- The premium is strong among stocks with low analyst coverage and high individual ownership.
- It rises monotonically with idiosyncratic volatility and with idiosyncratic volatility per investor.
- In the most recognition-poor groups, factor-model alphas are close to or above 1% per month.

Economic message. This is the cleanest evidence in favor of the Merton-style awareness story: media matters most where other information channels are weakest and undiversified idiosyncratic risk is most severe.

5.10 Table X: the media effect is not subsumed by other anomalies

Read:

- The no-media premium remains positive within analyst-dispersion groups and within high idiosyncratic-volatility groups.

Table X
Media Effect, Analyst Dispersion, and Idiosyncratic Volatility

This table examines whether the media effect is subsumed under the idiosyncratic volatility effect (Ang et al. (2006)) and the analyst dispersion effect (Diether, Malloy, and Scherbina (2002)). We double-sort stocks by media coverage and idiosyncratic volatility and analyst dispersion. Excess returns are computed using the DGTW characteristic-based benchmark methods. *t*-statistics are reported in parentheses.

Panel A: Double-Sort Media Coverage and Idiosyncratic Volatility				
	Low Idiosyncratic Volatility	Medium Idiosyncratic Volatility	High Idiosyncratic Volatility	High – Low Idiosyncratic Volatility
No media	0.001 (1.569)	0.002 (4.858)	0.005 (7.614)	0.005 (5.809)
Low media	0.002 (3.012)	0.001 (0.404)	0.003 (1.547)	0.001 (0.435)
High media	0.001 (2.215)	0.002 (1.907)	–0.005 (–2.405)	–0.006 (–3.713)
No – high media	0.001 (0.93)	0.000 (0.423)	0.010 (4.584)	
Panel B: Double-Sort Media Coverage and Analyst Dispersion				
	Low Dispersion	Medium Dispersion	High Dispersion	High – Low Dispersion
No media	0.006 (11.662)	0.004 (7.026)	–0.001 (–0.921)	–0.007 (–7.833)
Low media	0.003 (3.254)	0.004 (4.006)	–0.004 (–2.816)	–0.007 (–4.197)
High media	0.003 (3.297)	0.001 (1.219)	–0.004 (–2.557)	–0.006 (–4.039)
No – high media	0.004 (3.021)	0.003 (2.115)	0.003 (1.714)	

- The Ang et al. idiosyncratic-volatility puzzle appears mainly inside the high-media subset; among no-media stocks, high idiosyncratic volatility is associated with higher returns, which is closer to Merton’s prediction.
- Analyst dispersion remains a powerful separate effect, but it does not absorb the media effect.

Economic message. Media coverage is related to, but not reducible to, other information-based anomalies. It appears to be an additional dimension of the information environment.

6 Short summary

- The paper asks whether media coverage helps explain the cross-section of stock returns by broadening information dissemination and investor recognition.
- It measures firm-level media exposure using circulation-weighted article counts from four major U.S. newspapers for 1993–2002.
- Stocks with no media coverage outperform high-coverage stocks by 0.39% per month in raw returns and by about 0.23% per month after standard factor adjustments.
- The abnormal returns come mainly from neglected stocks earning positive alphas, not from well-covered stocks earning large negative alphas.
- The effect is strongest where investor recognition should matter most: small stocks, low-analyst stocks, stocks with high individual ownership, and stocks with high idiosyncratic volatility.
- The premium survives a long list of robustness checks and remains visible over long horizons, which makes a pure one-month reversal story unconvincing.
- The paper does not provide clean causal identification, but its strongest interpretation is that media coverage broadens recognition while illiquidity helps the pricing effect persist.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

The paper establishes a robust negative relation between mass-media coverage and future stock returns: stocks neglected by major national newspapers earn higher expected returns than heavily covered stocks, even after controlling for the standard factor structure of returns.

7.2 Economic intuition

Mass media reaches a wide set of investors. When a stock is broadly covered, more investors become aware of it, the shareholder base can diversify more easily, and the stock requires a lower expected return. When a stock remains in media “oblivion,” fewer investors recognize it, idiosyncratic risk is borne more narrowly, and the stock must offer a higher return. Trading frictions then help keep this premium from being instantaneously arbitrated away.

7.3 Takeaway

This paper’s most important message is that information dissemination has pricing consequences even when it is not about genuinely new information. Media visibility can matter for the cost of capital because it changes the breadth of investor awareness.